In [1]:
```
import pandas as pd
import numpy as np
from datetime import datetime
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("person.csv")
```

In [3]:
```
df.shape
```

Out[3]:
```
(17204, 18)
```

In [4]:
```
df.columns
```

Out[4]:
```
Index(['person_id', 'gender_concept_id', 'year_of_birth', 'month_of_birth',
       'day_of_birth', 'birth_datetime', 'race_concept_id',
       'ethnicity_concept_id', 'location_id', 'provider_id', 'care_site_id',
       'person_source_value', 'gender_source_value',
       'gender_source_concept_id', 'race_source_value',
       'race_source_concept_id', 'ethnicity_source_value',
       'ethnicity_source_concept_id'],
      dtype='str')
```

In [5]:
```
df.head(2)
```

Out[5]:

	person_id	gender_concept_id	year_of_birth	month_of_birth	day_of_birth	birth_date
0	1	8532	1959	1	4	1959-(00:(
1	2	8507	2005	9	21	2005-(00:(

In [6]:
```
print(df['gender_concept_id'].unique())
```

```
[8532 8507]
```

In [7]:
```
print(df['race_concept_id'].unique())
```

```
[8516 8527    0 8515]
```

In [8]:
```
print(df['ethnicity_concept_id'].unique())
```

```
[38003564 38003563]
```

In [9]:
```
print(df['year_of_birth'])
```

```
0        1959
1        2005
2        2023
3        1957
4        1953
         ...
17199    1966
17200    1973
17201    1932
17202    1970
17203    1994
Name: year_of_birth, Length: 17204, dtype: int64
```

In [10]:
```
current_year = datetime.now().year
```

In [11]:
```
df['age'] = current_year - df['year_of_birth']
```

In [12]:
```
print(df['age'])
```

```
0        67
1        21
2         3
3        69
4        73
         ..
17199    60
17200    53
17201    94
17202    56
17203    32
Name: age, Length: 17204, dtype: int64
```

In [13]:
```
print(df[['year_of_birth', 'age']])
```

```
       year_of_birth  age
0               1959   67
1               2005   21
2               2023    3
3               1957   69
4               1953   73
...              ...  ...
17199           1966   60
17200           1973   53
17201           1932   94
17202           1970   56
17203           1994   32

[17204 rows x 2 columns]
```

In [14]:
```
mean_age=df['age'].mean()
print (f'The mean of the patients age is {mean_age}')
```

```
The mean of the patients age is 44.9901185770751
```

In [15]:
```
median_age=df['age'].median()
print(f'The median of the patients age is{median_age}')
```

```
The median of the patients age is45.0
```

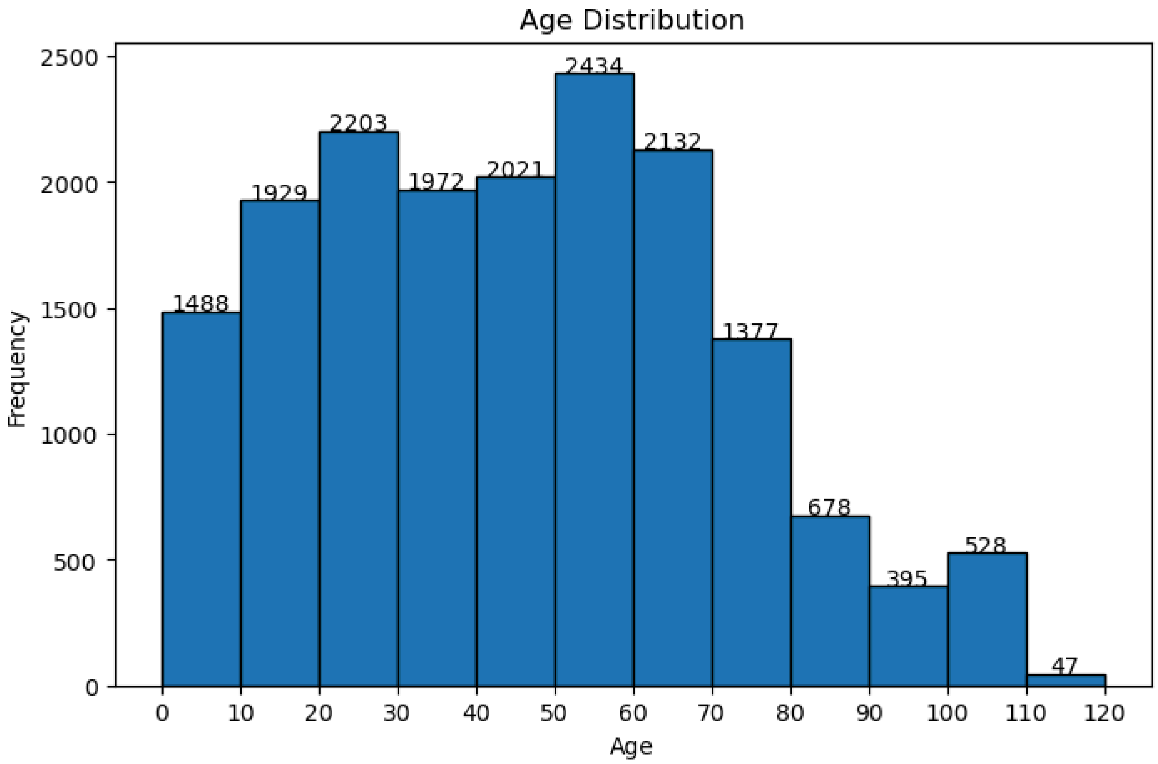
In [16]:
```
min_age=df['age'].min()
print(f'The minimum age among the Patients is {min_age}')
```

```
The minimum age among the Patients is 0
```

In [17]:
```
max_age=df['age'].max()
print(f'The maximum age among the patients is {max_age}')
```

```
The maximum age among the patients is 111
```

In [18]:
```
plt.figure(figsize=(8,5))
bins = range(0, int(df['age'].max()) + 10, 10)
counts, edges, patches = plt.hist(df['age'], bins=bins, edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.xticks(np.arange(0, int(df['age'].max()) + 10, 10))
plt.ylabel('Frequency')
plt.yticks(np.arange(0,3000,500))
for count, patch in zip(counts, patches):
    plt.text(patch.get_x() + patch.get_width() / 2,count,int(count), ha='center'
plt.grid(False)
plt.show()
```



In [19]:
```
dfc=pd.read_csv('concept.csv')
```

```
/tmp/ipykernel_118509/1735292881.py:1: DtypeWarning: Columns (0: standard_concep
t, 1: concept_code, 2: invalid_reason) have mixed types. Specify dtype option on
import or set low_memory=False.
  dfc=pd.read_csv('concept.csv')
```

In [20]:
```
dfc.columns
```

Out[20]:
```
Index(['concept_id', 'concept_name', 'domain_id', 'vocabulary_id',
       'concept_class_id', 'standard_concept', 'concept_code',
       'valid_start_date', 'valid_end_date', 'invalid_reason'],
      dtype='str')
```

In [21]:
```
dfc.shape
```

Out[21]: (2030608, 10)

In [22]:
```
dfc.head(5)
```

Out[22]:

	concept_id	concept_name	domain_id	vocabulary_id	concept_class_id	standard_con
0	3541502	Adverse reaction to drug primarily affecting t...	Condition	SNOMED	Disorder	I
1	3542356	Adverse reaction to other central nervous syst...	Condition	SNOMED	Disorder	I
2	42538812	Somatic hallucination	Condition	SNOMED	Clinical Finding	
3	40629514	Stillbirth	Condition	SNOMED	Clinical Finding	I
4	3539866	Adverse reaction to other skin, mucous membran...	Condition	SNOMED	Disorder	I

In [23]:
```
merged1_df = df.merge(dfc[['concept_id', 'concept_name']],left_on='gender_concep
```

In [24]:
```
merged1_df.head(2)
```

Out[24]:

	person_id	gender_concept_id	year_of_birth	month_of_birth	day_of_birth	birth_date
0	1	8532	1959	1	4	1959-(00:(
1	2	8507	2005	9	21	2005-(00:(

In [25]:
```
merged1_df = merged1_df.merge(
    dfc[['concept_id', 'concept_name']],
    left_on='race_concept_id',
    right_on='concept_id',
    how='left'
).rename(columns={'concept_name': 'race'}).drop(columns=['concept_id'])
```

In [26]:
```
merged1_df.head(2)
```

Out[26]:

	person_id	gender_concept_id	year_of_birth	month_of_birth	day_of_birth	birth_date
0	1	8532	1959	1	4	1959-(00:(
1	2	8507	2005	9	21	2005-(00:(

2 rows × 21 columns

In [27]:
```
merged1_df=merged1_df.merge(dfc[['concept_id', 'concept_name']], left_on = 'ethn
```

In [28]:
```
merged1_df.head(2)
```

Out[28]:

	person_id	gender_concept_id	year_of_birth	month_of_birth	day_of_birth	birth_date
0	1	8532	1959	1	4	1959-(00:(
1	2	8507	2005	9	21	2005-(00:(

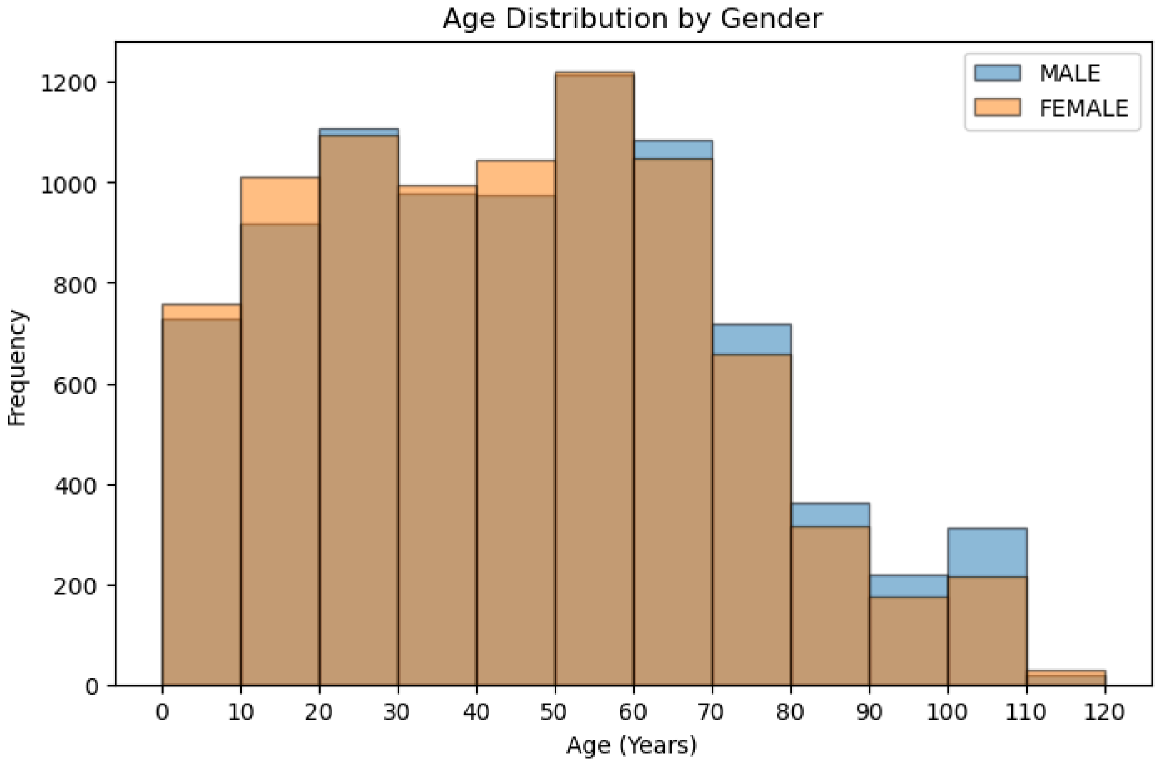
2 rows × 22 columns

In [29]:
```
crosstab = pd.crosstab(index=merged1_df['race'],columns=merged1_df['ethinicity']
```

In [30]:
```
print(crosstab)
```

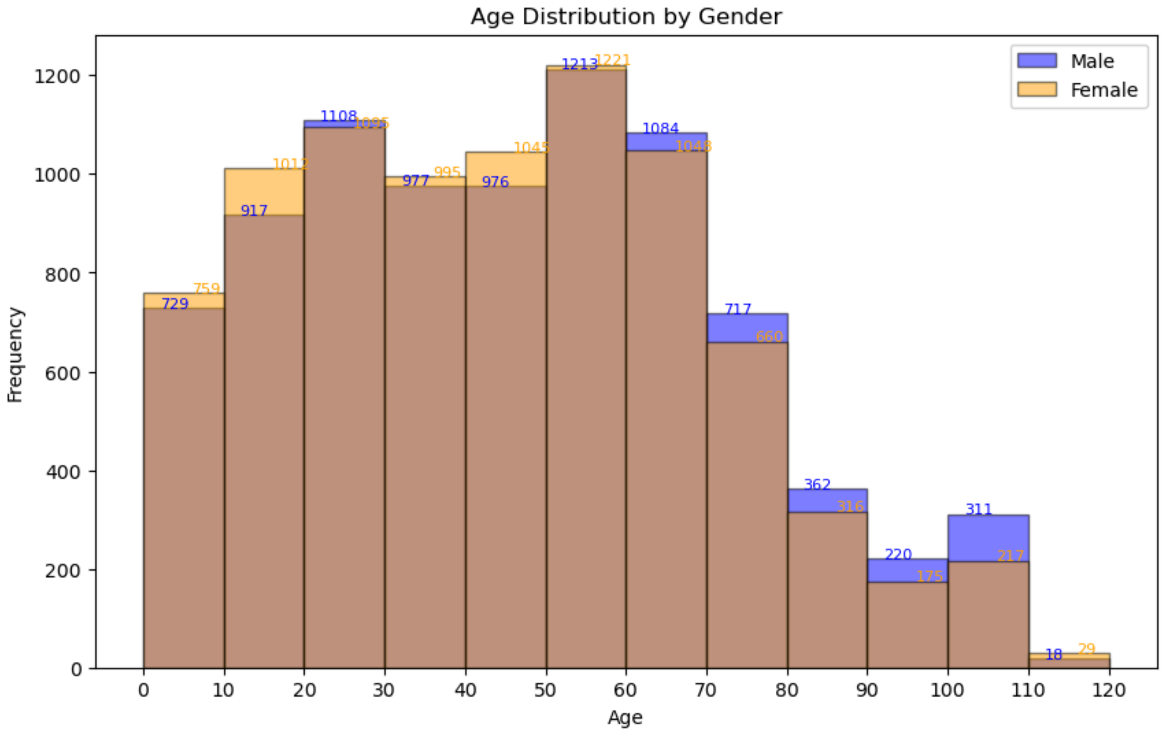
```
ethinicity                 Hispanic or Latino  Not Hispanic or Latino
race
Asian                                     113                    1032
Black or African American                 190                    1238
No matching concept                        67                     414
White                                    1454                   12696
```

In [31]:
```
plt.figure(figsize=(8,5))
bins = range(0, int(merged1_df['age'].max()) + 10, 10)
male = merged1_df[merged1_df['gender'] == 'MALE']['age']
female = merged1_df[merged1_df['gender'] == 'FEMALE']['age']
plt.hist(male, bins=bins, alpha=0.5, label='MALE', edgecolor='black')
plt.hist(female, bins=bins, alpha=0.5, label='FEMALE', edgecolor='black')
plt.xticks(np.arange(0, int(merged1_df['age'].max()) + 10, 10))
plt.xlabel('Age (Years)')
plt.ylabel('Frequency')
plt.title('Age Distribution by Gender')
plt.legend()
plt.show()
```



In [32]:

```
plt.figure(figsize=(10,6))
bins = range(0, int(merged1_df['age'].max()) + 10, 10)
male = merged1_df[merged1_df['gender'] == 'MALE']['age']
female = merged1_df[merged1_df['gender'] == 'FEMALE']['age']
counts_m, bins_m, _ = plt.hist(male, bins=bins, alpha=0.5,label='Male', color='b
counts_f, bins_f, _ = plt.hist(female, bins=bins, alpha=0.5,label='Female', colo
for i in range(len(counts_m)):
    plt.text(bins_m[i] + 2,counts_m[i],int(counts_m[i]),color='blue',fontsize=8)
    plt.text(bins_f[i] + 6,counts_f[i],int(counts_f[i]),color='orange',fontsize=
plt.xticks(np.arange(0, int(merged1_df['age'].max()) + 10, 10))
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Age Distribution by Gender')
plt.legend()
plt.show()
```


Age Distribution by Gender
Male
Female
Frequency
Age
1213
1221
1108
1084
1045
1012
995
977
976
917
759
729
717
660
362
316
311
220
217
175
18
29
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
100
110
120

```
In [ ]:
```

```
In [ ]:
```